

Final Project 1: Mind Track - Brain task app

1. Local Setup & Dockerization

- Cloning the source repository

```
MINGW64:/c/Users/Kousick Devaraj/Desktop/GuviProjects/Project1/Brain-Tasks-App

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1
$ git clone https://github.com/Vennilavanguvi/Brain-Tasks-App.git
Cloning into 'Brain-Tasks-App'...
remote: Enumerating objects: 11, done.
remote: Counting objects: 100% (2/2), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 11 (delta 0), reused 0 (delta 0), pack-reused 9 (from 1)
Receiving objects: 100% (11/11), 101.89 KiB | 7.28 MiB/s, done.

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1
$ ls -lrt
total 0
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 11 15:38 Brain-Tasks-App/

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1
$ cd Brain-Tasks-App/

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ ls -lrt
total 4
-rw-r--r-- 1 Kousick Devaraj 197121 1801 Sep 11 15:38 README.md
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 11 15:38 dist/

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$
```

- Creating Dockerfile

The application uses a lightweight nginx:alpine image from AWS ECR Public to serve static assets from the dist/ directory on port 80.

```
MINGW64:/c/Users/Kousick Devaraj/Desktop/GuviProjects/Project1/Brain-Tasks-App

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ ls -lrt
total 6
-rw-r--r-- 1 Kousick Devaraj 197121 1801 Sep 11 15:38 README.md
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 11 15:38 dist/
-rw-r--r-- 1 Kousick Devaraj 197121 368 Sep 11 15:54 Dockerfile_backup
-rw-r--r-- 1 Kousick Devaraj 197121 198 Sep 11 15:56 Dockerfile

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ cat Dockerfile
FROM nginx:alpine

# Copy pre-compiled dist folder to Nginx standard HTML root
COPY dist /usr/share/nginx/html

# Nginx listens on default HTTP port 80
EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```

Building local image:

Build local image

docker build -t brain-tasks-app:local .

```
kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ docker build -t brain-tasks-app:local .
[+] Building 19.0s (8/8) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 237B
=> [internal] load metadata for docker.io/library/nginx:alpine
=> [auth] library/nginx:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 318.06kB
=> [1/2] FROM docker.io/library/nginx:alpine@sha256:72ba65eb42c10344912a84ff42408db7d34f2feb642204570ab8fc5ffd29f1d3
=> resolve docker.io/library/nginx:alpine@sha256:72ba65eb42c10344912a84ff42408db7d34f2feb642204570ab8fc5ffd29f1d3
=> sha256:8f924cf5086c7b8a67e8b3e03b22851c1de8e11b0433c21904ef340d89e5014e 1.40kB / 1.40kB
=> sha256:6636b9fc203ce74d0b07a1d3f48cfa0151b23fa5e787232908e560673a66b553 20.52MB / 20.52MB
=> sha256:51900a10fb9cfad93aa74360e5f938efcbe6d3396a380867b7561c84e5cd2daf 1.21kB / 1.21kB
=> sha256:b93d6756163f687c200ed114df3904623bb05e7698b420efaf0ef3e81dffc 404B / 404B
=> sha256:58c524ea09ced2269cb14533d95b80de9fea9532abbf171585d8acee413893f2 956B / 956B
=> sha256:af7dd138f4591521064c208da6af07f487ebc3e5b350989c0ec2017b3e5208a3 628B / 628B
=> sha256:850bf2dcecf77924b238f0b7cbae7b436e5206453b784fe6b4adfc949add274a 4.40MB / 4.40MB
=> extracting sha256:850bf2dcecf77924b238f0b7cbae7b436e5206453b784fe6b4adfc949add274a
=> extracting sha256:af7dd138f4591521064c208da6af07f487ebc3e5b350989c0ec2017b3e5208a3
=> extracting sha256:58c524ea09ced2269cb14533d95b80de9fea9532abbf171585d8acee413893f2
=> extracting sha256:b93d6756163f687c200ed114df3904623bb05e7698b420efaf0ef3e81dffc
=> extracting sha256:51900a10fb9cfad93aa74360e5f938efcbe6d3396a380867b7561c84e5cd2daf
=> extracting sha256:8f924cf5086c7b8a67e8b3e03b22851c1de8e11b0433c21904ef340d89e5014e
=> extracting sha256:6636b9fc203ce74d0b07a1d3f48cfa0151b23fa5e787232908e560673a66b553
=> [2/2] COPY dist /usr/share/nginx/html
=> exporting to image
=> exporting layers
=> exporting manifest sha256:ab443c01d223c548d42f121e49813e505f81b697e5bf3a9b29bd3a91b384855
=> exporting config sha256:8bb771f1efcd652e9fe7f2ac05f68da40ee2f89e83a30bd35c4e6f5d7addb
=> exporting attestation manifest sha256:9d54f3227099bfff59cf21b1e22e9f23ee4156d2d727df314e54e0eb49caf6a7
=> exporting manifest list sha256:c324674683c0ccdafef06a7bd551542be7d5d5ac258b30560e20fc6656f857f1
=> naming to docker.io/library/brain-tasks-app:local
=> unpacking to docker.io/library/brain-tasks-app:local

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/niymhnvow7wy9k6e1wrhnktt
kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
```

Run container mapping host port 3000 to Nginx port 80
docker run -d -p 3000:80 --name brain-tasks-test brain-tasks-app:local

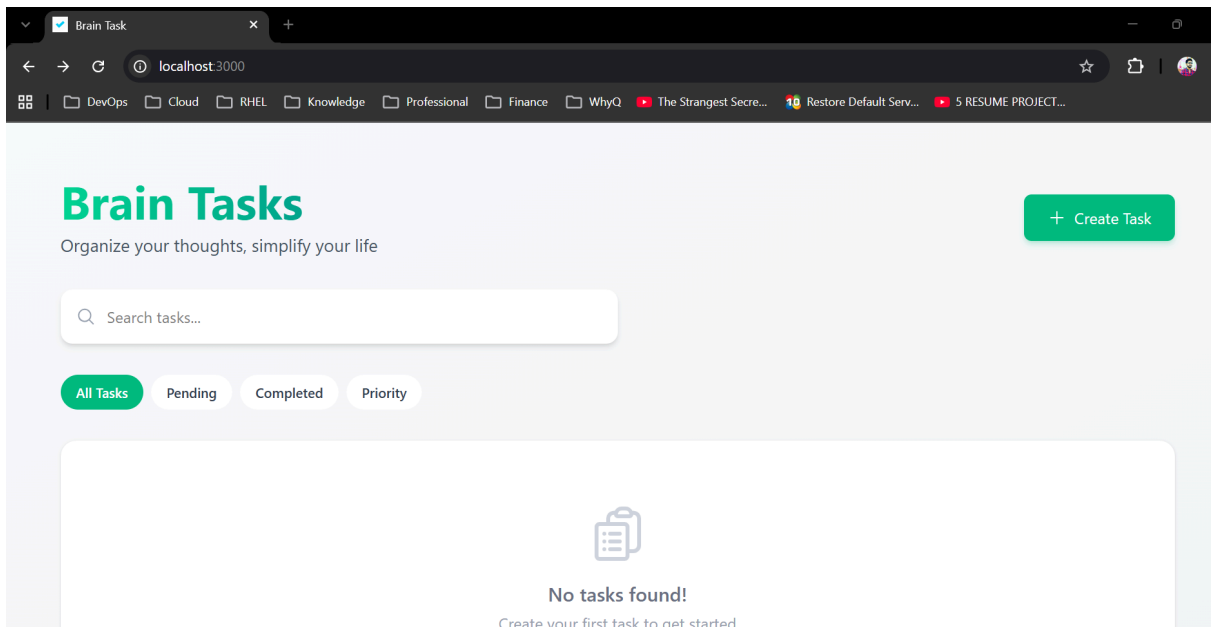
```
MINGW64:/c/Users/Kousick Devaraj/Desktop/GuviProjects/Project1/Brain-Tasks-App

kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ docker run -d -p 3000:80 --name brain-tasks-test brain-tasks-app:local
114c68b5f1056fb86b685a7103982578836d6382b41b4df8cd097b9313538c53

kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
114c68b5f105   brain-tasks-app:local   "/docker-entrypoint..." 6 seconds ago   Up 6 seconds   0.0.0.0:3000->80/tcp, [::]:3000->80/tcp   brain-tasks-test

kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
```

The app is validated on a local machine and it is working using container.



Create your own GitHub Repository & Push the Codebase

```
MINGW64~/c/Users/Kousick Devaraj/Desktop/GuviProjects/Project1/Brain-Tasks-App
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ ls -lrt
total 6
-rw-r--r-- 1 Kousick Devaraj 197121 1801 Sep 11 15:38 README.md
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 11 15:38 dist/
-rw-r--r-- 1 Kousick Devaraj 197121 368 Sep 11 15:54 Dockerfile_backup
-rw-r--r-- 1 Kousick Devaraj 197121 198 Sep 11 15:56 Dockerfile

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git remote add origin https://github.com/kousick-devaraj/Brain-tasks-app.git

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git branch -M main

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git status
On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Dockerfile
        Dockerfile_backup

nothing added to commit but untracked files present (use "git add" to track)

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git add Dockerfile
warning: in the working copy of 'Dockerfile', LF will be replaced by CRLF the next time Git touches it

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   Dockerfile

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Dockerfile_backup

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git commit -m "Initial commit with custom Dockerfile to create image for Brain task app"
[main 55ba793] Initial commit with custom Dockerfile to create image for Brain task app
1 file changed, 9 insertions(+)
create mode 100644 Dockerfile

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
```

```
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git push -u origin main
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 8 threads
Compressing objects: 100% (13/13), done.
Writing objects: 100% (14/14), 102.35 KiB | 51.18 MiB/s, done.
Total 14 (delta 0), reused 11 (delta 0), pack-reused 0 (from 0)
To https://github.com/kousick-devaraj/Brain-tasks-app.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$
```

2. AWS ECR Repository Creation

#Create a private container registry to store application images:

aws ecr create-repository --repository-name brain-tasks-app --region us-east-1

```

MINGW64/c/Users/Kousick Devaraj/Desktop/GuviProjects/Project1/Brain-Tasks-App
kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ aws ecr create-repository --repository-name brain-tasks-app --region us-east-1
{
  "repository": {
    "repositoryArn": "arn:aws:ecr:us-east-1::100000000000:repository/brain-tasks-app",
    "registryId": "786352305260",
    "repositoryName": "brain-tasks-app",
    "repositoryUri": "000000000000.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app",
    "createdAt": "2026-09-11T16:59:28.153000+05:30",
    "imageTagMutability": "MUTABLE",
    "imageScanningConfiguration": {
      "scanOnPush": false
    },
    "encryptionConfiguration": {
      "encryptionType": "AES256"
    }
  }
}

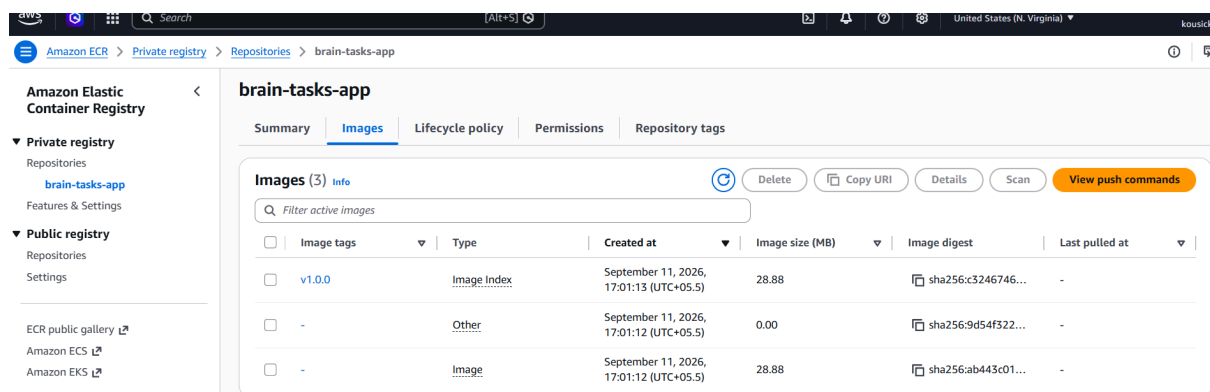
kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
114c68b5f105   brain-tasks-app:local   "/docker-entrypoint..." 59 minutes ago Up 59 minutes   0.0.0.0:3000->80/tcp, [::]:3000->80/tcp   brain-tasks-test

kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ docker tag brain-tasks-app:local 000000000000.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:v1.0.0

kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
114c68b5f105   brain-tasks-app:local   "/docker-entrypoint..." 59 minutes ago Up 59 minutes   0.0.0.0:3000->80/tcp, [::]:3000->80/tcp   brain-tasks-test

kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ docker push 000000000000.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:v1.0.0
The push refers to repository [000000000000.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app]
0fb4b85ff94: Pushed
44136fa355b3: Pushed
850bf2dcecff: Pushed
55afa1ecc21d: Pushed
a77dd138f459: Pushed
58c524ea09ce: Pushed
bc98d7675616: Pushed
51900e10fb9c: Pushed
6636b9fc203c: Pushed
8f924cf5086c: Pushed
d7c6b3982a5a: Pushed
v1.0.0: digest: sha256:c324674683c0ccdafef06a7bd551542be7d5d5ac258b30560e20fc6656f857f1 size: 856

```



3. AWS EKS Cluster Provisioning

Provision an EKS cluster with two worker nodes:

Bash

```

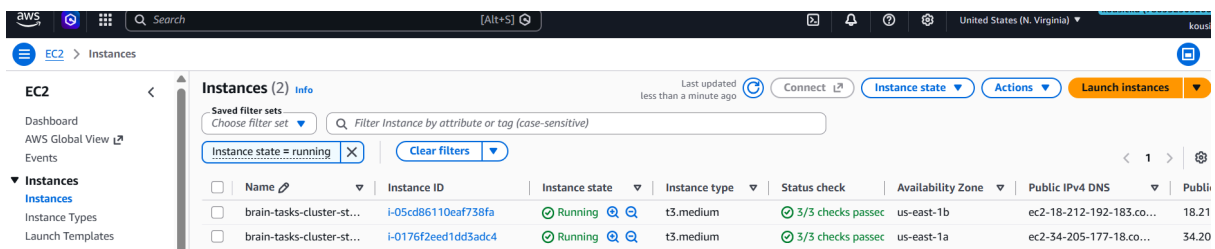
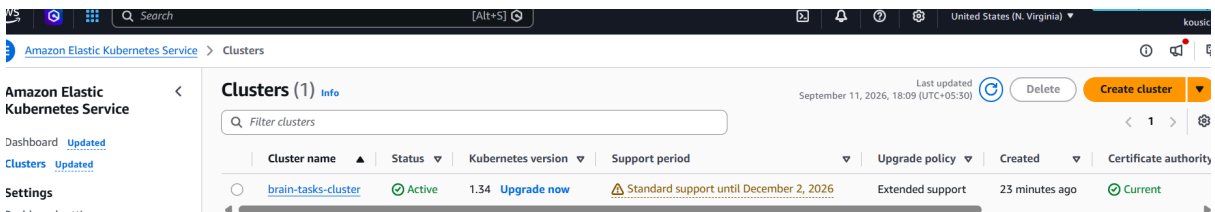
eksctl create cluster \
  --name brain-tasks-cluster \
  --region us-east-1 \
  --nodegroup-name standard-workers \
  --node-type t3.medium \
  --nodes 2 \
  --managed

```

Connect local kubectl context

```
aws eks update-kubeconfig --region us-east-1 --name brain-tasks-cluster
```

```
MINGW64/c:/Users/Kousick Devaraj/Desktop/GuviProjects/Project1/Brain-Tasks-App
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ eksctl create cluster --name brain-tasks-cluster --region us-east-1 --nodegroup-name standard-workers --node-type t3.medium --nodes 2 --managed
2026-09-11 17:46:02 [i] eksctl version 0.230.0
2026-09-11 17:46:02 [i] using region us-east-1
2026-09-11 17:46:03 [i] setting availability zones to [us-east-1a us-east-1b]
2026-09-11 17:46:03 [i] subnets for us-east-1a - public:192.168.0.0/19 private:192.168.64.0/19
2026-09-11 17:46:03 [i] subnets for us-east-1b - public:192.168.32.0/19 private:192.168.96.0/19
2026-09-11 17:46:03 [i] nodegroup "standard-workers" will use "" [AmazonLinux2023/1.34]
2026-09-11 17:46:03 [i] Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed networking add-ons will no longer be created by default. To maintain current behavior, explicitly set 'autoModeConfig.enabled: false' in your cluster configuration. Learn more: https://eksctl.io/usage/auto-mode/
2026-09-11 17:46:03 [i] using Kubernetes version 1.34
2026-09-11 17:46:03 [i] creating EKS cluster "brain-tasks-cluster" in "us-east-1" region with managed nodes
2026-09-11 17:46:03 [i] will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2026-09-11 17:46:03 [i] if you encounter any issues, check cloudFormation console or try 'eksctl utils describe-stacks --region=us-east-1 --cluster=brain-tasks-cluster'
2026-09-11 17:46:03 [i] Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "brain-tasks-cluster" in "us-east-1"
2026-09-11 17:46:03 [i] Cloudwatch logging will not be enabled for cluster "brain-tasks-cluster" in "us-east-1"
2026-09-11 17:46:03 [i] you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --region=us-east-1 --cluster=brain-tasks-cluster'
2026-09-11 17:46:03 [i] default addons metrics-server, vpc-cni, kube-proxy, coredns were not specified, will install them as EKS addons
2026-09-11 17:46:03 [i]
2 sequential tasks: { create cluster control plane "brain-tasks-cluster",
  2 sequential sub-tasks: {
    2 sequential sub-tasks: {
      1 task: { create addons },
      wait for control plane to become ready,
    },
    create managed nodegroup "standard-workers",
  },
}
2026-09-11 17:46:03 [i] building cluster stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:46:04 [i] deploying stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:46:04 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:47:05 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:48:05 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:49:07 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:50:08 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:51:10 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:52:10 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:53:11 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:54:13 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:55:14 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:56:15 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:57:16 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:58:17 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:58:23 [i] recommended policies were found for "vpc-cni" addon, but since OIDC is disabled on the cluster, eksctl cannot configure the requested permissions; the recommended way to provide IAM permissions for "vpc-cni" addon is via pod identity associations; after addon creation is completed, add all recommended policies to the config file, under "addon.PodIdentityAssociations", and run 'eksctl update addon'
2026-09-11 17:58:23 [i] recommended policies were found for "vpc-cni" addon, but since OIDC is disabled on the cluster, eksctl cannot configure the requested permissions; the recommended way to provide IAM permissions for "vpc-cni" addon is via pod identity associations; after addon creation is completed, add all recommended policies to the config file, under "addon.PodIdentityAssociations", and run 'eksctl update addon'
2026-09-11 17:58:23 [i] creating addon: vpc-cni
2026-09-11 17:58:24 [i] successfully created addon: vpc-cni
2026-09-11 17:58:24 [i] creating addon: kube-proxy
2026-09-11 17:58:25 [i] successfully created addon: kube-proxy
2026-09-11 17:58:25 [i] creating addon: coreDNS
2026-09-11 17:58:26 [i] successfully created addon: coreDNS
2026-09-11 18:00:32 [i] building managed nodegroup stack "eksctl-brain-tasks-cluster-nogroup-standard-workers"
2026-09-11 18:00:34 [i] deploying stack "eksctl-brain-tasks-cluster-nogroup-standard-workers"
2026-09-11 18:00:34 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-nogroup-standard-workers"
2026-09-11 18:01:06 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-nogroup-standard-workers"
2026-09-11 18:02:05 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-nogroup-standard-workers"
2026-09-11 18:03:03 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-nogroup-standard-workers"
2026-09-11 18:03:03 [i] waiting for the control plane to become ready
2026-09-11 18:03:07 [i] saved kubeconfig as "c:\Users\Kousick Devaraj\.kube\config"
2026-09-11 18:03:07 [i] no tasks
2026-09-11 18:03:07 [i] all EKS cluster resources for "brain-tasks-cluster" have been created
2026-09-11 18:03:09 [i] nodegroup "standard-workers" has 2 node(s)
2026-09-11 18:03:09 [i] node "ip-192-168-17-214.ec2.internal" is ready
2026-09-11 18:03:09 [i] node "ip-192-168-41-22.ec2.internal" is ready
2026-09-11 18:03:09 [i] waiting for at least 2 node(s) to become ready in "standard-workers"
2026-09-11 18:03:09 [i] nodegroup "standard-workers" has 2 node(s)
2026-09-11 18:03:09 [i] node "ip-192-168-17-214.ec2.internal" is ready
2026-09-11 18:03:09 [i] node "ip-192-168-41-22.ec2.internal" is ready
2026-09-11 18:03:09 [i] created 1 managed nodegroup(s) in cluster "brain-tasks-cluster"
2026-09-11 18:03:11 [i] creating addon: metrics-server
2026-09-11 18:03:11 [i] successfully created addon: metrics-server
2026-09-11 18:03:16 [i] kubectl command should work with "c:\Users\Kousick Devaraj\.kube\config", try 'kubectl get nodes'
2026-09-11 18:03:16 [i] EKS cluster "brain-tasks-cluster" in "us-east-1" region is ready
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
```



4. Kubernetes Manifests

- Creating the k8s deployment manifest and services.

```
MINGW64:/c:/Users/Kousick Devaraj/Desktop/GuviProjects/Project1/Brain-Tasks-App/k8s
$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: brain-tasks-deployment
  labels:
    app: brain-tasks-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: brain-tasks-app
  template:
    metadata:
      labels:
        app: brain-tasks-app
    spec:
      containers:
        - name: brain-tasks-container
          image: 786352305260.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest
          ports:
            - containerPort: 80
          resources:
            requests:
              cpu: "100m"
              memory: "128Mi"
            limits:
              cpu: "250m"
              memory: "256Mi"

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App/k8s (main)
$ cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: brain-tasks-service
spec:
  type: LoadBalancer
  ports:
    - name: http
      port: 3000
      targetPort: 80
      protocol: TCP
  selector:
    app: brain-tasks-app

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App/k8s (main)
$
```

Pushing data to ECR :

```
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ ls -lrt
total 10
-rw-r--r-- 1 Kousick Devaraj 197121 1801 Sep 11 15:38 README.md
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 11 15:38 dist/
-rw-r--r-- 1 Kousick Devaraj 197121 368 Sep 11 15:54 Dockerfile_backup
-rw-r--r-- 1 Kousick Devaraj 197121 198 Sep 11 15:56 Dockerfile
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 11 18:16 k8s/
-rw-r--r-- 1 Kousick Devaraj 197121 1533 Sep 11 18:29 buildspec.yml

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Dockerfile_backup
    buildspec.yml
    k8s/

nothing added to commit but untracked files present (use "git add" to track)

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git add k8s/ buildspec.yml
warning: in the working copy of 'buildspec.yml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'k8s/deployment.yaml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'k8s/service.yaml', LF will be replaced by CRLF the next time Git touches it

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   buildspec.yml
    new file:   k8s/deployment.yaml
    new file:   k8s/service.yaml

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Dockerfile_backup

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
```

```
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 1.52 KiB | 310.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kousick-devaraj/Brain-tasks-app.git
 55ba793..c122e4e  main -> main

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$
```

ECR Repo and image

Amazon ECR > Private registry > Repositories > brain-tasks-app

Amazon Elastic Container Registry

Private registry

Repositories

brain-tasks-app

Features & Settings

Public registry

Repositories

Settings

ECR public gallery

Amazon ECS

Amazon EKS

brain-tasks-app

Summary Images Lifecycle policy Permissions Repository tags

Images (3) Info

Filter active images

Image tags	Type	Created at	Image size (MB)	Image digest	Last pulled at
v1.0.0	Image Index	September 11, 2026, 17:01:13 (UTC+05.5)	28.88	sha256:c324674683c0ccd...	-
-	Other	September 11, 2026, 17:01:12 (UTC+05.5)	0.00	sha256:9d54f3227099bff...	-
-	Image	September 11, 2026, 17:01:12 (UTC+05.5)	28.88	sha256:ab443c01d223c54...	-

Buttons: Delete, Copy URI, Details, Scan, View push commands

Cluster info

Amazon Elastic Kubernetes Service > Clusters > brain-tasks-cluster

Cluster info

- Status: Active
- Kubernetes version: 1.34
- Support period: Standard support until December 2, 2026
- Provider: EKS
- Cluster health: 0
- Upgrade insights: 4
- Node health issues: 0
- Capability issues: 0

Details

- API server endpoint: <https://9727f12b511e0595ccb032bf97a0b0ff.gr7.us-east-1.eks.amazonaws.com>
- OpenID Connect provider URL: <https://oidc.eks.us-east-1.amazonaws.com/id/9727f12b511e0595ccb032bf97a0b0ff>
- Certificate authority: [View in IAM](#)
- Cluster IAM role ARN: [View in IAM](#)
- Created: 6 hours ago
- Cluster ARN: [View in IAM](#)
- Platform version: eks.32

5. IAM & EKS Authorization for CodeBuild

To authorize AWS CodeBuild to make deployment changes to EKS, map its IAM Role (CodeBuildEKSDeployRole) inside EKS:

Bash

```
eksctl create iamidentitymapping \
  --cluster brain-tasks-cluster \
  --region us-east-1 \
  --arn arn:aws:iam:::role/CodeBuildEKSDeployRole \
  --group system:masters \
  --username build
```

Identity and Access Management (IAM) > Roles > CodeBuildEKSDeployRole

Permissions policies (9)

You can attach up to 20 managed policies.

Policy name	Type	Attached entities
AmazonEC2ContainerRegistryPowerUser	AWS managed	2
AmazonEKSClusterPolicy	AWS managed	4
CloudWatchLogsFullAccess	AWS managed	2
CodeBuildBasePolicy-CodeBuildEKSDeployRole-us-east-1	Customer managed	1
CodeBuildCloudWatchLogsPolicy	Customer inline	0
CodeBuildCloudWatchLogsPolicy-MindTrack-us-east-1	Customer managed	1
CodeBuildCodeConnectionsSourceCredentialsPolicy-MindTrack-us-east-1	Customer managed	1
CodeBuildEKSDeployPolicy	Customer inline	0
CodeBuildVpcPolicy-MindTrack-us-east-1	Customer managed	1

6. Automated CI/CD Pipeline Configuration

The pipeline triggers automatically on every git push to main.

- Source:** GitHub via AWS CodeConnections.
- Build:** AWS CodeBuild compiles Docker images, tags them with commit hashes, pushes to ECR, and applies Kubernetes manifests (kubectl apply).

3. Deploy: Managed inside CodeBuild using cluster credentials.

The screenshot shows the AWS CodeBuild console for the 'MindTrack' project. The left sidebar contains navigation links for Developer Tools, CodeBuild, and Build projects. The main content area displays the project configuration and a table of build history.

Configuration

- Source provider: GitHub
- Primary repository: kousick-devaraj/Brain-tasks-app
- Artifacts upload location: -
- Service role: arn:aws:iam::786352305260:role/service-role/CodeBuildEKSDeployRole
- Public builds: Disabled

Build history

Build run	Status	Build number	Source version	Submitter	Duration	Completed
MindTrack:6560f8-3319-4d5a-9ed2-a35a4667a465	Succeeded	18	arn:aws:s3:::codepipeline-us-east-1-d0aa71b62a42-41be-b555-d4b24b0a0c58/brain-tasks-pipeline/SourceArti/ARlieAw	codepipeline/brain-tasks-pipeline	31 seconds	23 minutes ago
MindTrack:3703e8dc-07eb-49a9-9e4f-ed3f5c990b0e	Failed	17	arn:aws:s3:::codepipeline-us-east-1-d0aa71b62a42-41be-b555-d4b24b0a0c58/brain-tasks-pipeline/SourceArti/QbVVD6g	codepipeline/brain-tasks-pipeline	38 seconds	27 minutes ago

Pipeline

The screenshot shows the AWS CodePipeline console for the 'brain-tasks-pipeline'. The pipeline is in progress, with the 'Source' stage completed and the 'Build' stage in progress.

brain-tasks-pipeline

Source (13607a79-0265-4495-bf9c-63b30f3d451d)

- All actions succeeded.
- Source: GitHub (via GitHub App)
- c122e4e9 Source: Adding K8s manifests and CodeBuild specification

Build (13607a79-0265-4495-bf9c-63b30f3d451d)

- In progress: 1
- Build: AWS CodeBuild
- c122e4e9 Source: Adding K8s manifests and CodeBuild specification

The screenshot shows the AWS CodePipeline console for the 'brain-tasks-pipeline'. The pipeline is now completed, with both the 'Source' and 'Build' stages showing success.

brain-tasks-pipeline

Source (67458b29-9cc6-40f5-aa7f-fb6442c984ed)

- All actions succeeded.
- Source: GitHub (via GitHub App)
- efa7fec2 Source: removed conflict markers

Build (67458b29-9cc6-40f5-aa7f-fb6442c984ed)

- All actions succeeded.
- Build: AWS CodeBuild
- efa7fec2 Source: removed conflict markers

7. Monitoring & Operational Logs

Build execution and container deployment events are streamed to CloudWatch Logs under the `/aws/codebuild/brain-tasks-build` log group.

The screenshot shows the AWS CodeBuild console for a specific build. The build status is 'Succeeded'. The build number is 18. The build started on Sep 11, 2026 at 11:13 PM (UTC+5:30). The build ended at the same time. The build ARN is `arn:aws:codebuild:us-east-1:786352305260:build/MindTrack:e6560fc8-3319-4d3a-9ed2-a35a4667a465`. The resolved source version is `efa7fec2642c83b5379b810036ef17d5486a5aec`. The build logs are visible, showing the last 225 lines of the build log. The logs indicate that the build is running on CodeBuild On-demand, and the build is successful.

The screenshot shows the AWS CodeBuild console for the same build, displaying the build logs. The logs show the build process, including the download of source code, the execution of buildspec.yml, and the deployment of the application to Amazon EKS. The logs indicate that the build is successful, and the application is deployed to the EKS cluster.

Cloud watch logs

The screenshot shows the AWS CloudWatch console for the `/aws/codebuild/MindTrack` log group. The log streams are listed, showing the build ID, the build name, and the last event time. The log streams are sorted by the last event time, with the most recent log stream at the top.



Verification & Live Application

To fetch the dynamic AWS LoadBalancer URL:

```
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Downloads
$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
brain-tasks-deployment-7c98b7bc6b-mj8tr  1/1     Running   0           4m27s
brain-tasks-deployment-7c98b7bc6b-qh6d1  1/1     Running   0           4m26s

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Downloads
$ kubectl get all
NAME                                READY   STATUS    RESTARTS   AGE
pod/brain-tasks-deployment-7c98b7bc6b-mj8tr  1/1     Running   0           5m7s
pod/brain-tasks-deployment-7c98b7bc6b-qh6d1  1/1     Running   0           5m6s

NAME                                TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
service/brain-tasks-service          LoadBalancer  10.100.206.51    a87010d6dcd614b718c9e322d9740e23-1814529202.us-east-1.elb.amazonaws.com  3000:30699/TCP  5m7s
service/kubernetes                    ClusterIP      10.100.0.1      <none>           443/TCP          5h25m

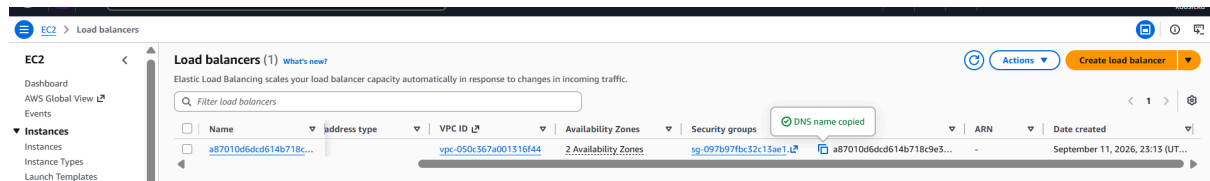
NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/brain-tasks-deployment  2/2     2             2           5m8s

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/brain-tasks-deployment-7c98b7bc6b  2         2         2       5m9s

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Downloads
```

LB url -

<http://a87010d6dcd614b718c9e322d9740e23-1814529202.us-east-1.elb.amazonaws.com:3000/>



Validating output using LoadBalancer URL of the k8s cluster.

